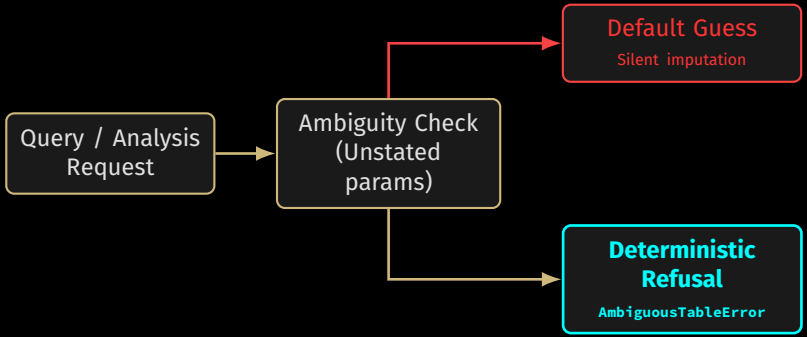
jnwb: Dataset-Agnostic Ephys

Contract-Driven Tooling for Neurophysiologists and AI Agents

Hamed Nejat
Bastos Lab · Vanderbilt University

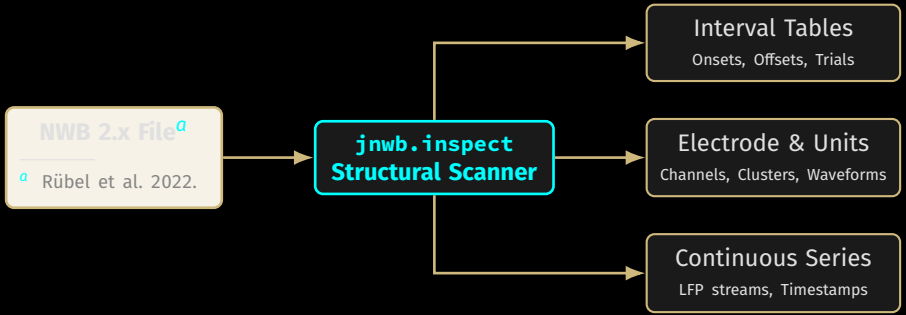
October 29, 2026

Refusal Over Guessing: Explicit Failures vs. Silent Guesses



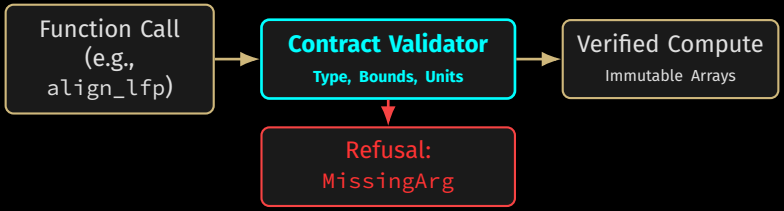
Silent fallback masks scientific divergence; strict refusal enables reproducible automation.

Dynamic Layout Discovery: Read, Never Assume



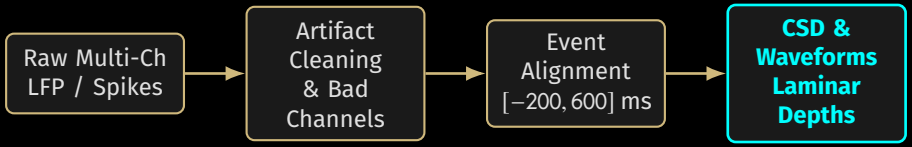
Discovered metadata binds downstream execution without hardcoded coordinate conventions.

Scientific Parameter Contracts: Rigorous Signatures



Explicit alignment window, bin width, and reference scheme required on every call.

Laminar Signal Pipeline: Continuous to Structured Arrays



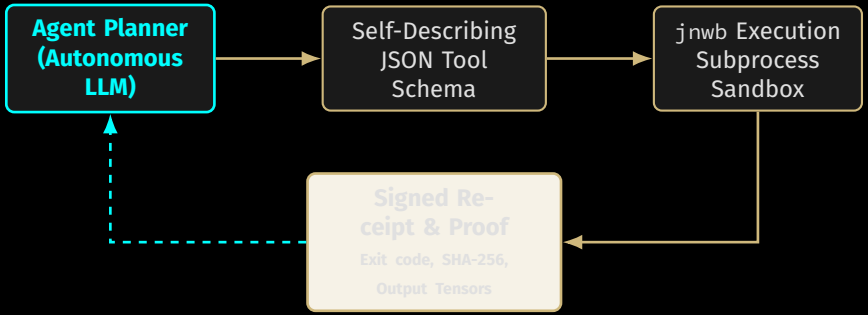
Input Guarantee

No silent baseline shifting or unrecorded filtering.

Output Guarantee

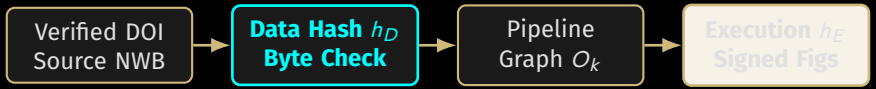
Dimension-checked tensors ready for laminar analysis.

Agent-Assisted Architecture: Autonomous Pair Programming



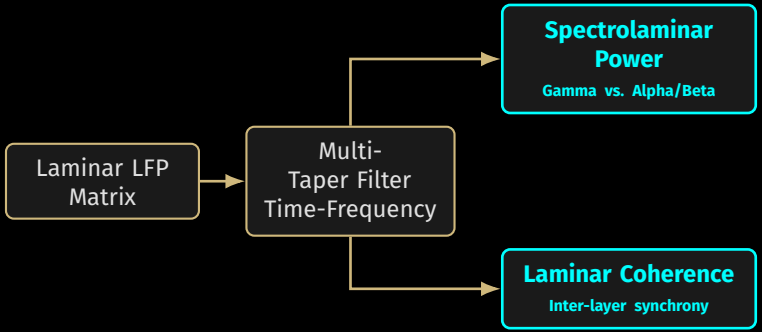
Agents act via structured tools; verification gates prevent model hallucinations.

Provenance Locking: Full Reproducibility Ledger



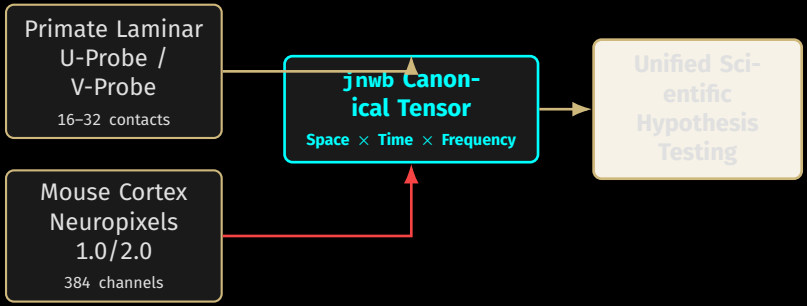
Every figure in a manuscript traces back to raw bytes through an unbroken mathematical receipt.

Spectral & Coherence Engine: Spectrolaminar Metrics



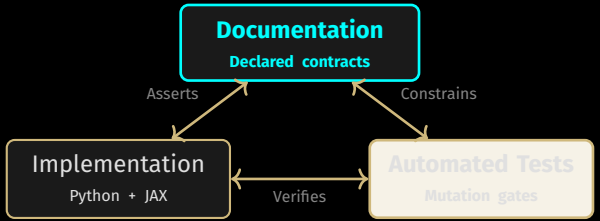
Automated identification of laminar layer boundaries from spectral power crossovers.

Cross-Platform Generality: Species and Hardware Agnostic



Tool abstractions eliminate custom per-lab scraping code and disparate data conventions.

The Tripartite Loop: Mutually Constraining Architecture



A change in one vertex triggers automatic gate failure in the other two.

Summary: Verifiable Electrophysiology Pipelines



For Neurophysiologists

Transparent analyses with clear error boundaries.

For Agent Systems

Strict contract validation that eliminates silent hallucinations.